ZHONGYI (JAMES) GUO

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EDUCATION

MS, Epidemiology and Clinical Research, Stanford University (**GPA: 3.86/4.00**) 06/2025
BS, Biometry & Statistics *and* **Biological Sciences,** Cornell University (**GPA: 3.57/4.30**) 05/2023
Honors: *cum laude*, Dean's List

WORK EXPERIENCE

Biostatistics Intern, Corcept Therapeutics 06/2026 – (est.) 08/2026
- Matched 120+ external controls from RWD using propensity score, emulating index dates and analyzing outcome with Kaplan-Meier curves, log-rank tests, and Cox proportional hazards models.
- Developed three-tier strategies to guide the venture capital arm's setup, analyzing SEC filings for fund size and constructed portfolios with real-world companies across stages and deal sizes.
- Built and validated patient profiles across ~20 panels in Medidata Clinical Data Studio, mapping data from annotated CRFs and creating derived variables to streamline clinical trial data review.

Research Associate, the Corces Lab, Gladstone Institutes 07/2025 – 04/2026
- Pioneered an R package implementing a permutation-test pipeline to reduce false positives in scRNA-seq differential gene expression analysis of Alzheimer's disease to identify therapeutic targets, integrating ten public *Synapse* large-scale datasets to demonstrate its real-world use.
- Applied AlphaGenome to study how genetic variants of the *MAPT* gene alter splicing patterns.
- Merged 250 samples' arrow files, scored and removed doublets, and performed dimensionality reduction (UMAP & iterative LSI) on scATAC-seq data using ArchR.
- Communicated statistical findings and visualizations using story-telling to diverse audiences.

Graduate Student Researcher, the Kundaje Lab, Stanford University 04/2024 – 06/2025
- Preprocessed and integrated high-throughput scRNA-seq and scATAC-seq data of psychiatric disorders, applying summary statistics, quality control, feature engineering, and dimension reduction.
- Performed differential gene expression and peak accessibility analyses at pseudobulk (using Wald test and LRT in *DESeq2*) and single-cell levels (using Wilcoxon rank-sum test and MAST in Seurat).
- Trained and interpreted cell type-specific ChromBPNET, a CNN-based DL model, to assess effects of potential genetic disease variants on transcription factor binding and chromatin accessibility.
- Delivered clear updates to professors and colleagues regularly, simplifying complex statistics.

Graduate Student Researcher, the Graff Lab, UCSF (remote) 10/2023 – 06/2025
- Applied t-test, PCA, PLS-DA, random forest, logistic & linear regression, chemical similarity enrichment analysis, pathway analysis, and WGCNA to LC-MS metabolomics data, characterizing metabolites/biomarkers associated with Black-White prostate cancer disparities.
- Interpreted statistical findings and communicated them through clear scientific writing and effective data visualizations; defended before Stanford Epidemiology faculty and students.

TECHNICAL SUMMARY

Languages: R, Python, SAS (Base certified), SQL, LaTeX, HTML, CSS, Javascript (D3.js)
Tools: Git, GitHub, Conda, RStudio, Jupyter Notebook, High Performance Computing, MS Office
Core Skills: statistical computing, quantitative modeling, data-driven & cross-functional collaboration
Aesthetics: Adobe (Illustrator, Photoshop, After Effect), Procreate.

Last Updated on 07/21/2026

PUBLICATION

[1] (Submitted to Nature Neuroscience, 2025) Shin, J.‡, Brady, E.‡, Chen, C., Lauderdale, K., Agrawal, A., Zhang, Y., Jiang, X., Nambiar, P., Herbert, J., Mallen, D., Ly, K., **Guo, Z.**, Sant, C., Thomas, R., Miller, S., Cobos, I., Palop, J.. APOE4 and Aβ synergize to drive neuronal network dysfunction and lysosomal-ER proteostasis dysregulation in the preclinical stages of Alzheimer's disease.

[2] (Submitted to Cell, 2026) Bendall, S., Ee, R., Amouzgar, M., Afaghani, J., Vijayaragavan, K., Cannon, B., Mrdjen, D., Tebaykin, D., Spence, A., Sant, C., Aley, D., **Guo, Z.**, Sedov, K., Zafar, F., Montine, K., Perna, A., Serrano, G., Beach, T., Angelo, M., Schüle, B., Corces, M.R., Montine, T.. APOE4 Drives Uniquely Dysfunctional Human Microglial States in Alzheimer's Disease.

Manuscripts in preparation:

[1] (In Progress; Aiming for BMC Medicine) **Guo, Z.‡**, Chen, D.‡, Stopsack, K. H., Soule, P., Ajit, D., Ramamoorthy, P., Hoffmann, T. J., Chan, J. M., Mucci L. A., Graff, R. E.. Metabolomic Disparities Between Black and White Men with Metastatic Hormone-Sensitive Prostate Cancer: A Pilot Study.

[2] (In Progress; Aiming for Cell) Qu, P., Wang, T., Jessa, S., **Guo, Z.**, Guo, H., Purmann, C., Monte, E., Jiang, L., Yang, X., Zhou, B., Kundu, S., Kundaje, A., Wong, W., Hallmayer, J. F., Urban, A. E., Snyder, M. P.. Multi-modal functional genomics analysis of bipolar disorder and schizophrenia. (Title is tentative.)

[3] (In Progress; Aiming for Nature Neuroscience) Sant, C., **Guo, Z.**, Corces, M. R.. Preventing false discoveries in Alzheimer's disease single-cell sequencing data using permutation testing. (Title is tentative.)

‡ indicates co-first authorship.